Author: In principle, the concept of the **instantaneous velocity** (or in other words, "velocity at a given moment of time") must be independent of the measurement accuracy. The velocity you are talking about, that is, the ratio $\frac{\Delta s}{\Delta t}$ is nothing more than the **average velocity** during Δt. It is not the instantaneous velocity at all. Of course, you are right when you say that the smaller is Δt the closer is the value of the average velocity to the value of the instantaneous velocity. However, no matter how small is Δt, the ratio $\frac{\Delta s}{\Delta t}$ is always only the average velocity during Δt.

Reader: Then a better definition of the instantaneous velocity is beyond me.

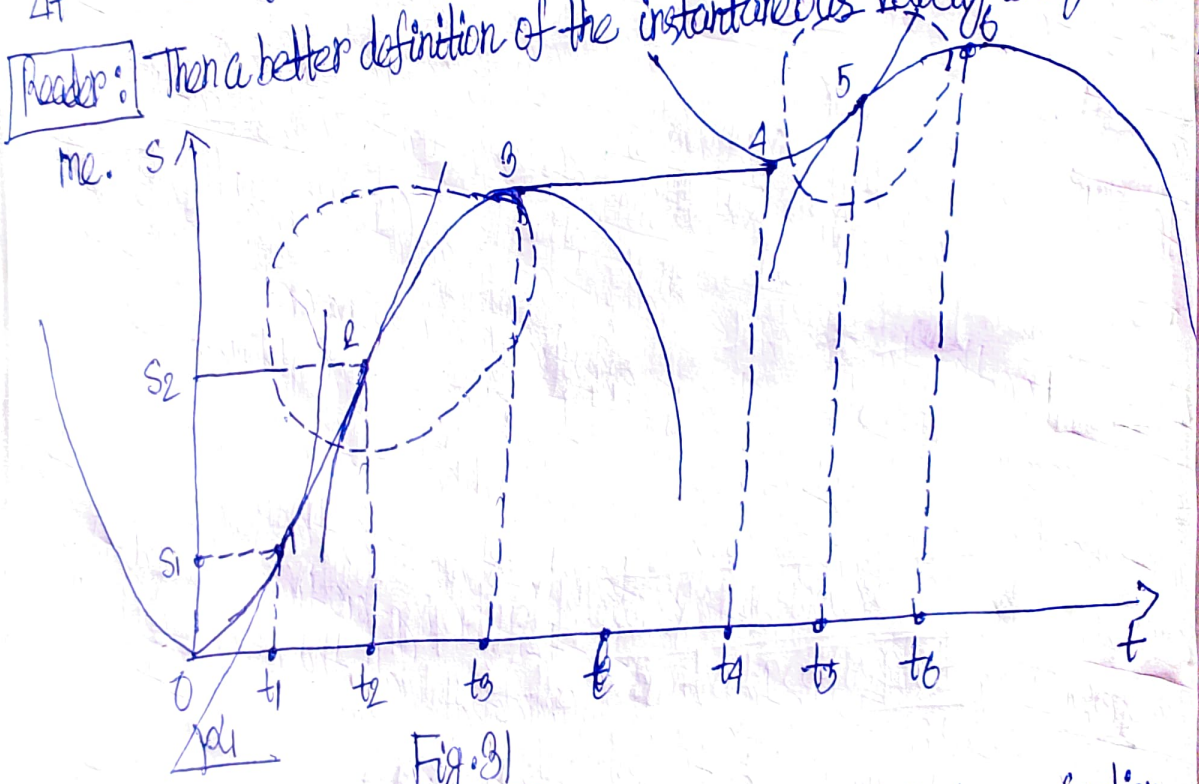

Fig. 31

Author: Consider a graph of distance covered by a body plotted as a function of time, that is, the graph of the function $s=s(t)$. This graph is shown in Fig. 31 by a solid line. Note that in physics one typically uses the same symbol to denote both a function and its values (in this case we use the symbol s).

Reader: The figure also shows several thin lines.

Author: The thin lines (parabolas and straight lines) are shown only to indicate how the graph of $s=s(t)$ was plotted. This graph is thus composed of "pieces" of parabolas and straight lines. For instance, for the time interval from 0 to t_1 the graph is represented by a "piece" of